

Sprint 7 Review Report

Project: Music Course Platform

Sprint: Sprint 7 (Week 14–15)

Date: April 27, 2026

Scrum Master: Liu Lu

Team Members: Luo Ying, Lu Liu, Su Wai Phyo, Yicheng Chen

Sprint Goal

Ensure the system is fully functional, usable, and aligned with the Product Owner's requirements by conducting comprehensive functional and non-functional testing, validating system performance and usability, and finalizing documentation for project delivery.

Key Objectives

- Create a detailed test plan including objectives, resources, test environment, and test scope
- Perform final unit testing after code refactoring from Sprint 6
- Identify, track, and resolve bugs using a structured bug tracking table
- Execute CI/CD pipeline using Jenkins with successful Docker deployment
- Generate static code analysis report using SonarQube
- Ensure code quality meets required standards (Grade A or at least B)
- Conduct heuristic evaluation based on usability principles
- Perform User Acceptance Testing (UAT) based on Sprint 6 criteria
- Review and refine user stories in Trello
- Update technical documentation and reflect system changes

Completed Tasks

Test Plan Creation – Completed by Luo Ying and Su Wai Phyo

A comprehensive test plan was created, including objectives, test environment, and both functional and non-functional testing scope.

Unit Testing – Completed by Su Wai Phyo and Lu Liu

Unit tests were executed across core modules such as authentication, API endpoints, and database operations to ensure correct functionality.

Bug Tracking and Fixing – Completed by All Members

All identified issues were recorded in a structured bug tracking table and resolved to ensure system stability.

CI/CD Pipeline Execution – Completed by Yicheng Chen

The Jenkins pipeline was successfully executed, including project build and Docker deployment.

SonarQube Analysis – Completed by Yicheng Chen

Static code analysis was performed, confirming acceptable code quality, low duplication, and manageable complexity.

Performance Testing – Completed by Su Wai Phyoe and Yicheng Chen

System responsiveness was evaluated, and database queries were optimized to improve performance.

Heuristic Evaluation – Completed by Luo Ying and Yicheng Chen

Usability testing was conducted based on standard principles, resulting in improved UI consistency and user feedback.

User Acceptance Testing (UAT) – Completed by Luo Ying and Su Wai Phyoe

System functionality was validated against user requirements and acceptance criteria.

Documentation Update – Completed by Lu Liu

Technical documentation and README were updated to reflect testing results and system improvements.

User Stories Review – Completed by Lu Liu

User stories were reviewed and refined in Trello to ensure alignment with the final system functionality.

Team Contribution Summary

Team Member	Assigned Tasks	Time Spent (hrs)	In-class Tasks
Luo Ying	Test plan, UAT, heuristic evaluation	12 hrs	Submitted
Lu Liu	Documentation update, user stories refinement	13 hrs	Submitted
Su Wai Phyoe	Unit testing, bug fixing, performance testing	13 hrs	Submitted
Yicheng Chen	CI/CD pipeline, SonarQube, deployment	14 hrs	Submitted

Testing Summary

Testing activities covered both functional and non-functional aspects of the system.

Functional Testing:

All core features were validated through unit testing, and system behavior matched expected results.

Non-functional Testing:

Performance testing confirmed acceptable response time under normal usage. Heuristic evaluation improved usability and interface consistency. UAT confirmed that the system meets user requirements.

Code Quality:

SonarQube analysis showed acceptable quality level (Grade A or B), with no critical vulnerabilities detected.

Challenges and Solutions

Bug Management Complexity

Resolved by introducing a structured bug tracking table to efficiently record and fix issues.

Performance Issues Under Load

Improved by optimizing database queries and reducing unnecessary processing.

Usability Inconsistencies

Addressed through heuristic evaluation and UI refinements.

CI/CD Integration Issues

Solved by adjusting pipeline configurations and ensuring stable Docker deployment.

Conclusion

Sprint 7 successfully validated the system through comprehensive testing and quality assurance processes. All functional and non-functional requirements were met, and the system is stable, usable, and ready for final delivery. Testing, evaluation, and documentation updates were completed in alignment with project requirements.

Trello Board : <https://trello.com/b/6EcNAcQ8/sep1musiccourseplatform-sprint7>

GitHub Link : <https://github.com/chenyicheng1998/MusicCoursePlatform>